

Course Code: 205-01

Course Title: Digital Logic and Computer Organization

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Course Title	Digital Logic and Computer Organization										
Credits	4										
Course Category	Major Course										
Level of Course	200-299 (Intermediate Level)										
Teaching per Week	4 Hours										
Minimum weeks per Semester	15 (Including class work, examination, preparation etc.)										
Review / Revision	-										
Implementation Year:	A.Y. 2024-2025										
Purpose of Course	The purpose of the "Digital Logic and Computer Organization" course is to provide students with a fundamental understanding of digital systems and computer architecture. Through the study of digital logic principles, students learn to design and analyze basic digital circuits and logic gates. Additionally, the course covers computer organization concepts, including CPU architecture, memory systems, and input/output devices, preparing students for further study in computer science and related fields. By mastering these foundational concepts, students develop the skills necessary for hardware design, computer engineering, and software development roles in the technology industry.										
Course Objective	1. Understand digital logic fundamentals, including Boolean algebra and logic circuits. 2. Explore computer organization principles, focusing on CPU, memory, and input/output devices. 3. Master number systems and data representation techniques. 4. Analyze and simplify logic circuits using methods like Karnaugh maps and Boolean algebra. 5. Gain insight into memory organization, storage systems, and memory management techniques.										
Pre-requisite	Basic knowledge of Computer										
Course Outcomes	CO1 : Demonstrate understanding of Boolean algebra and logic gates. CO2 : Design and analyze combinational and sequential logic circuits. CO3 : Explain CPU architecture, memory systems, and I/O devices. CO4 : Implement memory management techniques and cache memory organization. CO5 : Analyze and optimize computer architecture performance. CO6 : Evaluate emerging trends in digital systems and computer organization.										
Mapping between Course Outcomes(CO) with Program Specific Outcomes(PSO)			PSO 1	PSO2	PSO 3	PSO 4	PSO 5	PSO 6	PSO 7	PSO 8	
	CO1										
	CO2										
	CO3										
	CO4										
	CO5										
	CO6										
Course Content	Unit 1: Introduction to Digital Logic 1.1 Basic Concepts 1.1.1 Digital vs. Analog 1.1.2 Binary systems and Boolean algebra 1.2 Logic Gates 1.2.1 AND, OR, NOT, NAND, NOR, XOR, XNOR gates 1.2.2 Gate-level minimization 1.3 Boolean Algebra and Simplification										

	1.3.1 Boolean functions 1.3.2 Simplification using Karnaugh Maps (K-Maps) 1.3.3 Quine-McCluskey method 1.4 Activities: 1.4.1 Practice problems on Boolean algebra simplification 1.4.2 Hands-on lab session with logic gate simulators
	Unit 2: Combinational Circuits 2.1 Introduction to Combinational Circuits 2.1.1 Definition and examples 2.2 Design and Analysis of Combinational Circuits 2.2.1 Adder, Subtractor (Half and Full) 2.2.2 Multiplexer and Demultiplexer 2.2.3 Encoder and Decoder 2.3 Comparators and Parity Generators 2.3.1 Design and implementation of comparators 2.3.2 Parity generator and checker circuits
	Unit 3: Sequential Circuits 3.1 Introduction to Sequential Circuits 3.1.1 Difference between combinational and sequential circuits 3.1.2 Basic memory elements: Latches and Flip-Flops 3.2 Types of Flip-Flops** 3.2.1 SR, D, JK, T flip-flops 3.2.2 Triggering mechanisms: edge-triggered and level-triggered 3.3 Design of Sequential Circuits** 3.3.1 Counters (synchronous and asynchronous) 3.3.2 Shift registers
	Unit 4: Computer Organization Basics 4.1 Introduction to Computer Organization 4.1.1 Structure and components of a computer system 4.1.2 Von Neumann architecture 4.2 Central Processing Unit (CPU) 4.2.1 CPU organization 4.2.2 Instruction set architecture (ISA) 4.2.3 Instruction cycle and execution 4.3 Memory Hierarchy - Types of memory: primary vs. secondary - Cache memory: principles and organization
	Unit 5: Input/Output and Storage Systems 5.1. I/O Systems 5.1.1 I/O interface and ports 5.1.2 Types of I/O devices 5.2 Storage Systems 5.2.1 Hard disk drives (HDDs) and solid-state drives (SSDs) 5.2.2 Optical and magnetic storage 5.3 Introduction to Buses 5.3.1 Bus architecture and types 5.3.2 Data transfer techniques: programmed I/O, interrupt-driven I/O, direct memory access (DMA)

Reference Books	1. "Digital Design: Principles and Practices," John F. Wakerly, Pearson, ISBN: 9780134460093 2. "Computer Organization and Design: The Hardware/Software Interface," David A. Patterson and John L. Hennessy, Morgan Kaufmann, ISBN: 9780123747501 3. "Digital Electronics: Principles and Applications," Roger L. Tokheim, McGraw-Hill Education, ISBN: 9780073510880 4. "Fundamentals of Digital Logic with Verilog Design," Stephen Brown and Zvonko Vranesic, McGraw-Hill Education, ISBN: 9780073380544 5. "Structured Computer Organization," Andrew S. Tanenbaum, Pearson, ISBN: 9780132916523 6. "Digital Systems: Principles and Applications," Ronald J. Tocci, Neal S. Widmer, and Gregory L. Moss, Pearson, ISBN: 9780135103821 7. "Computer Organization and Architecture: Designing for Performance," William Stallings, Pearson, ISBN: 9780134101613 8. "Introduction to Logic Design," Alan B. Marcovitz, McGraw-Hill Education, ISBN: 9780073191645 9. "Computer Organization and Architecture," Barry B. Brey, Pearson, ISBN: 9780132936330 10. "Logic and Computer Design Fundamentals," M. Morris Mano and Charles R. Kime, Pearson, ISBN: 9780133760637
Teaching Methodology	Class Work, Discussion, Lab work, Self-Study, Seminars and/or Assignments
Evaluation Method	50% Internal assessment. 50% External assessment.